Medical Insurance Charges Analysis

Understanding Demographics and Smoking behaviours with Statistical Analysis

Jialun Darren Huang

Introduction

- Medical insurance helps individuals manage financial burden of healthcare expenses
- Insurance charges can differ with individual demographics and lifestyle characteristics.
- Understanding these differences can help us gain insights to:
  - *What are some factors that associate to higher medical insurance costs*
  - *Insurers and policymakers: Design a fairer, unbiased, and more data driven insurance pricing model*
  - *Individuals with medical insurance: How their own lifestyle can impact medical and their insurance costs, so they can choose better lifestyle choices*

Problem Statement

- This presentation investigates whether smoking behaviour is associated with demographic factors and higher medical insurance charges

- Specifically, we look into:

  - *Whether Sex and region are associated with smoking status*
  - *Whether smokers have a higher medical insurance charges, compared to non-smokers*

- These relationships would be analysed using

  - *Categorical association with Chi-square tests*
  - *Hypothesis testing, comparing differences in medical insurance charges between groups*

Data

- Medical Insurance Cost Dataset from Kaggle by user Mosap Abdel-Ghany (See references).
- Dataset contains individual medical insurance records with demographic and lifestyle information
- No explicit sampling method was said in the data source, however, it could have been sampled (e.g. from a registry or larger population), as the data contains just over 1000 insurance records.
- Subset of features was selected for the analysis: sex, region, smoker, charges. (See Table 1 for Dataset Description)
- Conducted preprocessing steps to ensure quality of data and its integrity
  - *Checked for any missing values in the features*
  - *Checked for any incorrect or inconsistent values (e.g. no negative values in Medical insurance charges)*
  - *Converted categorical features into factor format in R for better interpretability within R*

```
colSums(is.na(df))
```

```
##  smoker     sex  region charges 
##       0       0       0       0
```

- No missing values in the dataset

Data

Table 1. Dataset Description

Feature	Description	Data.Type.and.Values
smoker	Smoking Status of the individual	Nominal Categorical, yes/no
sex	Sex of the individual	Nominal Categorical, female/male
region	Region of the individual	Nominal Categorical, southwest/southeast/northwest/northeast
charges	Medical insurance charges of the individual	Discrete Numerical, Positive Integers

```
summary(df$charges)
```

```
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     1122    4740    9382   13270   16640   63770
```

- Medical insurance charges do not have any negative values, showing no error in the data
- Will dive deeper into their distribution in the later slides

Descriptive Statistics and Visualisation (Categorical Features)

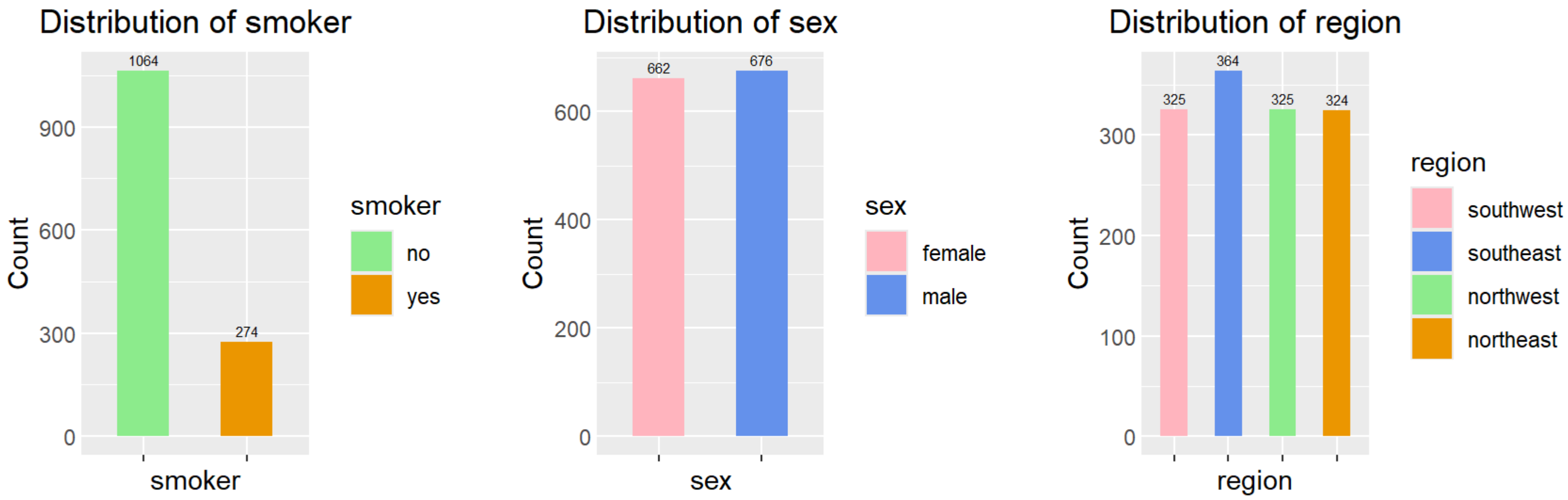

Fig 1. Distribution for Categorical Features

- Distribution of sex and region and quite balanced across the different categories
- More non-smokers than smokers within the dataset
- Imbalance may reflect a more realistic population behaviour, where smoking is less common due to its known negative health impacts

Descriptive Statistics and Visualisation (Numerical Features)

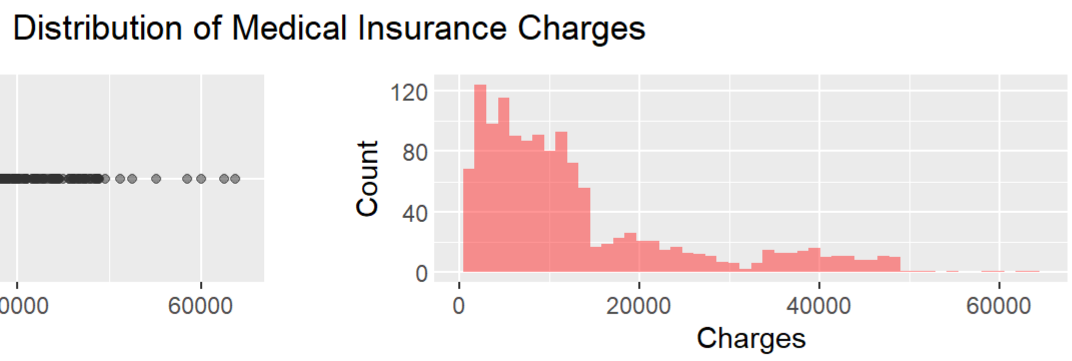

Fig 2. Distribution of Medical Insurance Charges

Table 2. Descriptive Statistics for Medical Insurance Charges

Minimum	Q1	Median	Q3	Max	IQR	Mean	SD
1121.874	4740.287	9382.033	16639.91	63770.43	11899.63	13270.42	12110.01

- Distribution is very right skewed, with a small percentage of individuals having very high medical insurance charges
- Majority of individuals have medical insurance charges around 4740 to 16639
- The standard deviation is larger at 12110 than IQR 11899, meaning that the variability is also affected by high medical insurance charges

Categorical association (between Region and Sex)

- Chi-square test of independence analysis to examine whether there is an association between sex and region, if the distribution of males and females differ across the different regions
- Identify if the demographic structure is consistent across regions, testing for biases
- H0: Sex and region are independent (No association)
- HA: Sex and region are associated
- Significance level: alpha = 0.05

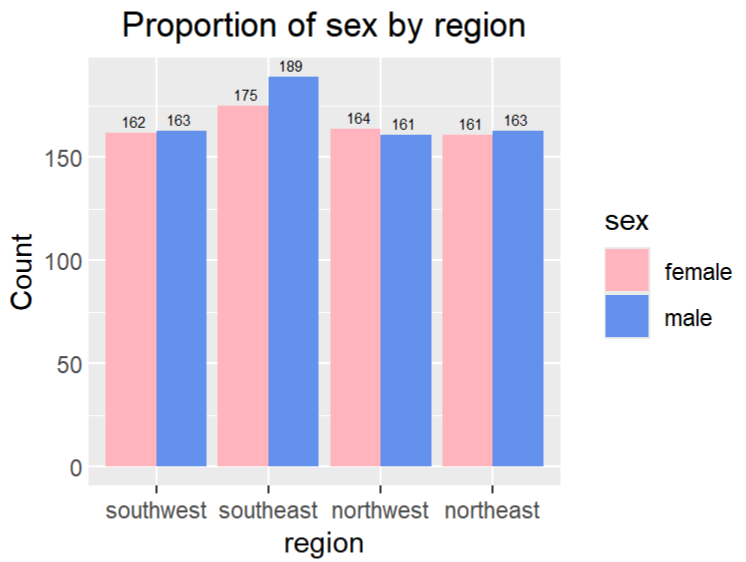

Fig 3. Proportion of Sex by Region

- Visually, it does not look like there is a difference in sex across regions

Categorical association (between Region and Sex)

Table 3. Contigency Table of Sex by Region. Observed vs (Expected)

region	female	male
southwest	162 (160.8)	163 (164.2)
southeast	175 (180.096)	189 (183.904)
northwest	164 (160.8)	161 (164.2)
northeast	161 (160.305)	163 (163.695)

```
##
##  Pearson's Chi-squared test
##
## data:  table(df[[feature_1]], df[[feature_2]])
## X-squared = 0.43514, df = 3, p-value = 0.9329
```

- Contingency table:
  - *Met assumptions where no more than 25% of cells have expected counts < 5*
  - *showed observed frequencies close to expected frequencies (in brackets)*
- Chi-square test showed p-value = 0.93, which is greater than signifance level at 0.05.
- Fail to reject null hypothesis, indicating there is no statisitcally significant association between sex and region

Categorical association (between Smoker Status and Sex)

- Chi-square test of independence analysis to examine whether there is an association between sex and smoking status, if the distribution of males and females differ across the different smoking statuses
- Determine if smoking behaviour differs between males and females
- H0: Sex and smoking status are independent (No association)
- HA: Sex and smoking status are associated
- Significance level: alpha = 0.05

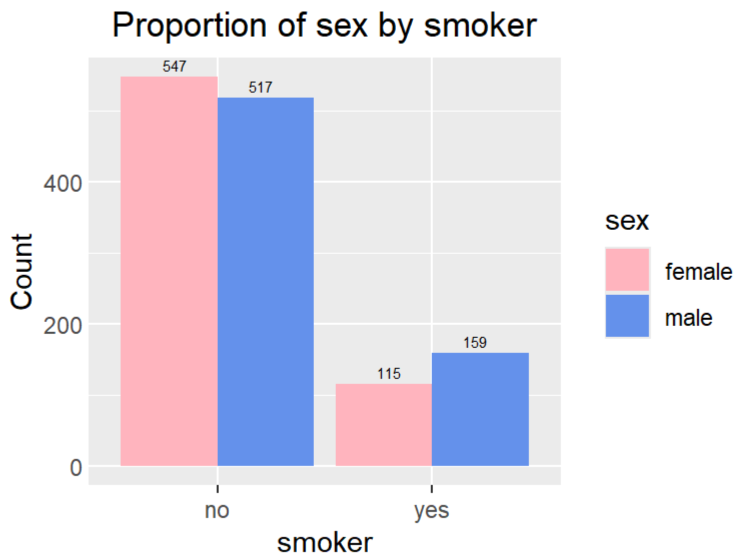

Fig 4. Proportion of Sex by Smoking Status

- Visible difference, where males are more likely to smoke than females

Categorical association (between Smoker Status and Sex)

Table 4. Contigency Table of Sex by Smoking Status Observed vs (Expected)

smoker	female	male
no	547 (526.433)	517 (537.567)
yes	115 (135.567)	159 (138.433)

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  table(df[[feature_1]], df[[feature_2]])
## X-squared = 7.3929, df = 1, p-value = 0.006548
```

- Contingency table:
  - *Met assumptions where no more than 25% of cells have expected counts < 5*
  - *showed observed frequencies are slightly further apart to expected frequencies (in brackets)*
  - *Females were observed to smoke less than expected*
  - *Male were observed to smoke more than expected*
- Chi-square test showed p-value = 0.006, which is less than significance level at 0.05. Reject null hypothesis, indicating there is a statisitcally significant association between sex and smoking status. Males are more likely to smoke than Females

Hypothesis Testing (between Medical Insurance Charges and Smoking)

- Independent two-sample T-test analysis if medical insurance charges differs between smoking status
- Identify if smoking is associated with higher medical insurance costs
- H0: Mean medical insurance charges are equal for all smoking groups
- HA: Mean medical insurance charges differ between smokers and non-smokers
- Significance level: alpha = 0.05

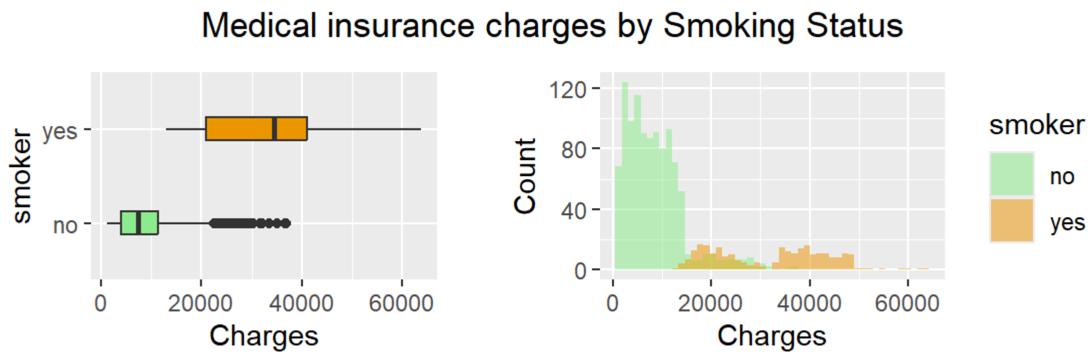

Fig 5. Distribution of Medical Insurance Charges by Smoking Status

- Visible difference, where smokers have a higher medical insurance charge

Hypothesis Testing (between Medical Insurance Charges and Smoking)

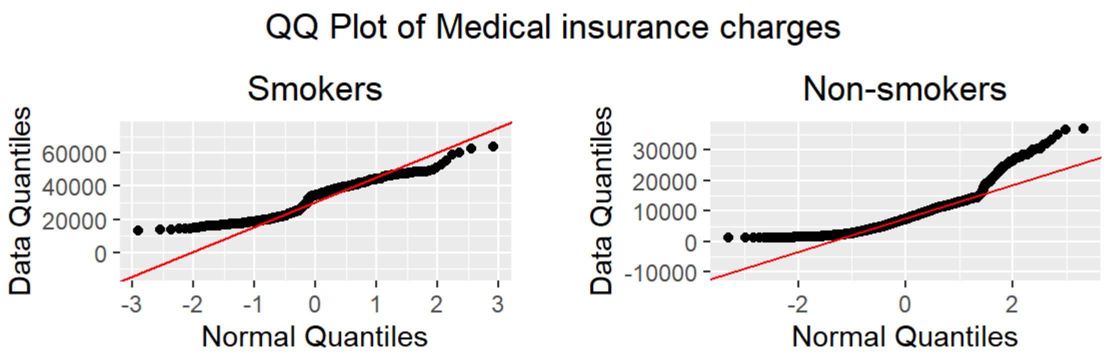

Fig 6. Normality Test with QQ Plot for Medical Insurance Charges by Smoking Status

Table 5. Homogeneity of Variance Test for Medical Insurance Charges by Smoking Status

	Df	F value	Pr(>F)
group	1	332.6135	1.559328e-66
	1336	NA	NA

- Assumptions:
  - *Normality test with Q-Q plot showed the data is not normally distributed, however since both smoking groups had sample sizes > 30, T-test still remains valid under Central Limit Theorem*
  - *Homogeneity of variance test showed uneual variances as the p-value < 0.05*
    - Instead of conducting the standard T-test, we will have to conduct the Welch two-sample T-test instead

Hypothesis Testing (between Medical Insurance Charges and Smoking)

```
t.test(
  charges ~ smoker, data = df,
  var.equal = FALSE, mu = 0, alternative = "two.sided"
)
```

```
##
##  Welch Two Sample t-test
##
## data:  charges by smoker
## t = -32.752, df = 311.85, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group no and group yes is not equal to 0
## 95 percent confidence interval:
##  -25034.71 -22197.21
## sample estimates:
##  mean in group no mean in group yes
##          8434.268         32050.232
```

- Welch T-test showed p-value = 2.2e-16, which is less than significance level. Reject null hypothesis, indicating there is a statistically significant difference in mean medical insurance charges between smokers and non-smokers
- Smokers incur significantly higher medical insurance charges than non-smokers, suggesting that smoking might have an association with increased medical costs

Hypothesis Testing (between Medical Insurance Charges and Sex)

- Independent two-sample T-test analysis if medical insurance charges differs between sex
- Identify if the different sex is associated with higher medical insurance costs
- H0: Mean medical insurance charges are equal for all sexes
- HA: Mean medical insurance charges differ between sexes
- Significance level: alpha = 0.05

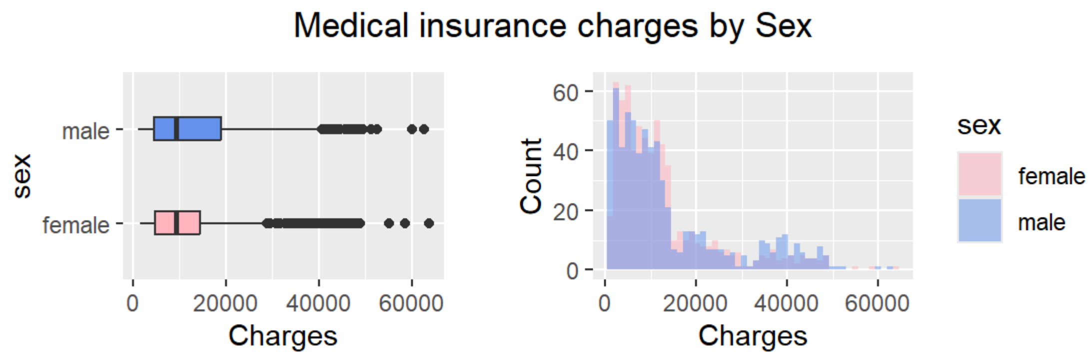

Fig 7. Distribution of Medical Insurance Charges by Sex

- Not much visible difference, however the Q1-Q3 range for medical insurance charges is larger for Male than Female

Hypothesis Testing (between Medical Insurance Charges and Sex)

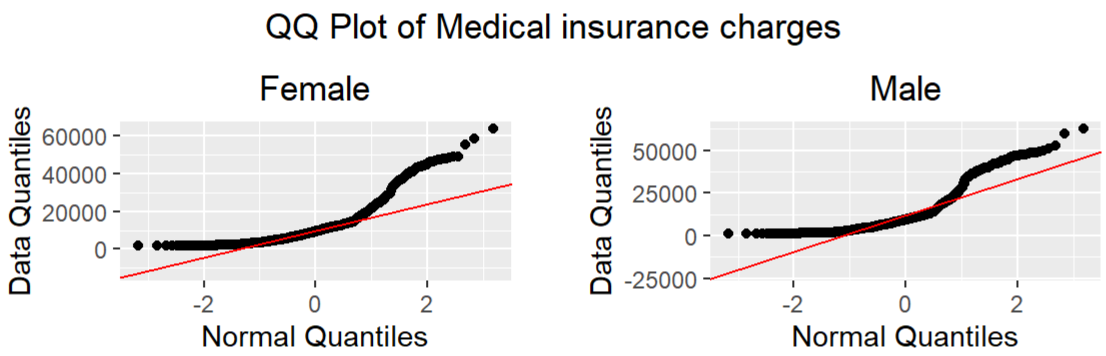

Fig 8. Normality Test with QQ Plot for Medical Insurance Charges by Sex

Table 6. Homogeneity of Variance Test for Medical Insurance Charges by Sex

	Df	F value	Pr(>F)
group	1	9.909251	0.001680877
	1336	NA	NA

- Assumptions:
  - *Normality test with Q-Q plot showed the data is not normally distributed, however since both sexes had sample sizes > 30, T-test still remains valid under Central Limit Theorem*
  - *Homogeneity of variance test showed uneual variances as the p-value < 0.05*
    - Instead of conducting the standard T-test, we will have to conduct the Welch two-sample T-test instead

Hypothesis Testing (between Medical Insurance Charges and Sex)

```r
t.test(
  charges ~ sex, data = df,
  var.equal = FALSE, mu = 0, alternative = "two.sided"
)
```

```
##
##  Welch Two Sample t-test
##
## data:  charges by sex
## t = -2.1009, df = 1313.4, p-value = 0.03584
## alternative hypothesis: true difference in means between group female and group male is not equal to 0
## 95 percent confidence interval:
##  -2682.48932   -91.85535
## sample estimates:
## mean in group female   mean in group male
##             12569.58             13956.75
```

- Welch T-test showed p-value = 0.03584, which is less than significance level. Reject null hypothesis, indicating there is a statistically significant difference in mean medical insurance charges between sexes
- Males incur significantly higher medical insurance charges than Females, although this difference might be partly explained through other factors such as the higher proportion of smokers for males.

Discussion

- Findings:
  - *Dataset is unbiased in terms of data collected between sex and region*
  - *Significant association was found between sex and smoking status, where males are more likely to smoke than females*
  - *Smokers had significantly higher medical insurance charges than non-smokers*
  - *While on average that males also had a higher medical insurance charge than females, it can be due to the difference explained between the higher proportion of smokers among males*
- Strengths and Limitations:
  - *Combined categorical association and hypothesis testing to investigate a couple of perspectives on how factors relate to medical insurance charges*
  - *Hypothesis stated and assumption checks are always performed before conducting statistical tests*
  - *Results are also supported with statistical analysis and graphical visualisations*
  - *However, only selected features were analysed, while other factors (e.g. age and BMI) where not considered*
  - *Regression analysis not conducted, lose information such as having a combination of multiple features that significantly affect medical insurance charges*

Conclusion

- Regardless of region, smoking appears to play a big role in explaning the difference in medical insurance charges, with more smokers being males than females

- Smoking behaviour could be a key factor in higher medical costs, highlighting the importance of how lifestyle choices can influence in medical expenses and insurance outcomes

- Health organisations can explore ways and interventions to help reduce smoking among both males and females, which may help reduce overall medical insurance charges

- Further Investigation:

  - *Perform multiple regression analysis to determine if smoking status is still a significant feature affecting charges, or a combination of other features (inclusive/exclusive of smoking status)*

  - *Investigate more features on the individual's demographic and lifestyle choices*

References

1. Mosap Abdel-Ghany. (2025). Medical Insurance Cost Dataset. Kaggle. https://www.kaggle.com/datasets/mosapabdelghany/medical-insurance-cost-dataset
2. Hansjörg Neth. (2026). D.3 Basic R colors https://bookdown.org/hneth/ds4psy/D.3-apx-colors-basics.html
3. Yihui Xie, J.J. Allaire, Garrett Grolemund. (2026). 4.2 Slidy presentation. https://pkg.yihui.org/rmarkdown-book/slidy-presentation